



1h 29m
left



Help

All

1

2

3

4

Problem Statement

The Paris Olympics are here and the Olympics committee has decided to introduce a new game. The game is as follows

- There are n ordered hurdles and athlete can cross max k consecutive hurdles. For example, if there are 3 hurdles and $k = 1$, athlete must remove either the first two hurdles, the last two hurdles or the first and last hurdle.
- Each hurdle is associated with a score and athlete can decide which hurdles to remove to achieve maximum score.
- The Athlete wants your help in identifying max score possible for given n , k , and provided hurdle scores.
- **Note:** Hurdles are ordered and cannot be reordered.

Input Format

- First line contains two space separated integers n (number of hurdles) and k (maximum no of consecutive hurdles that athlete can cross)
- Each line i of the n subsequent lines contains an integer denoting the score of hurdle i where $0 \leq i < n$

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq n$
- $0 \leq \text{score of a hurdle} \leq 2 \cdot 10^9$

Output Format

- Print a single integer denoting the max score that athlete can achieve

Assembler 32 (Mars 4.5)

```
1  .data
2  userInput: .space 20 # reserve space to store user's input
3
4  .text
5  main:
6
7  # getting user's input
8  li $v0, 8
9  la $a0, userInput
10 li $a1, 20
11 syscall
12
13 # display user's input on console
14 li $v0, 4
15 la $a0, userInput
16 syscall
17
18 # code for program end
19 li $v0, 10
20 syscall
21
```

Test Results Custom Input

ASUS



1h 29m
left



Help

All

1

2

3

4

denoting the score of hurdle i where $0 \leq i < n$

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq n$
- $0 \leq \text{score of a hurdle} \leq 2 \cdot 10^9$

Output Format

- Print a single integer denoting the max score that athlete can achieve

Sample Input 0

6 2

1

2

3

1

6

10

Sample Output 0

21

Explanation 0

There are 6 hurdles and athlete can keep max 2 consecutive hurdles. Athlete can remove first and fourth hurdle leading to configuration 2 3 _ 6 10 and a score of 21.

Sample Input 1

5 4

1

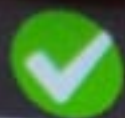
2

Assembler 32 (Mars 4.5)

```
1  .data
2  userInput: .space 20 # reserve space to store user's input
3
4  .text
5  main:
6
7  # getting user's input
8  li $v0, 8
9  la $a0, userInput
10 li $a1, 20
11 syscall
12
13 # display user's input on console
14 li $v0, 4
15 la $a0, userInput
16 syscall
17
18 # code for program end
19 li $v0, 10
20 syscall
21
```

Test Results Custom Input

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1h 29m
left



3
1
6
10

Sample Output 0

21

Help

All

1

2

3

4

Explanation 0

There are 6 hurdles and athlete can keep max 2 consecutive hurdles.
Athlete can remove first and fourth hurdle leading to configuration
2 3 _ 6 10 and a score of 21.

Sample Input 1

5 4
1
2
3
4
5

Sample Output 1

14

Explanation 1

There are 5 hurdles and athlete can keep max 4 consecutive hurdles.
Athlete can remove first hurdle leading to configuration _ 2 3 4 5
and a score of 14.

Execution time limit
10 seconds

Assembler 32 (Mars 4.5)

```
1 .data
2 userInput: .space 20 # reserve space to store user's input
3
4 .text
5 main:
6
7 # getting user's input
8 li $v0, 8
9 la $a0, userInput
10 li $a1, 20
11 syscall
12
13 # display user's input on console
14 li $v0, 4
15 la $a0, userInput
16 syscall
17
18 # code for program end
19 li $v0, 10
20 syscall
21
```

Test Results Custom Input

ASUS



1h 28m
left



Help

All

2. Help Mike Get Rich

Coding

UNSOLVED

Problem Statement

Mike Ross is a brilliant lawyer, and wins all his cases. But he is not a morning person.

His law firm brought up a new rule, wherein to be eligible for a bonus, a lawyer must go through one pending case doc every day if that streak is broken, no bonus for that year.

To keep the case distribution fair, few rules were laid out.

1. Whenever a lawyer enters the office, he/she would wait in line for the next case distribution event.
2. Every case distribution event in a day has a fixed number of cases.
3. At the distribution time, lawyers with earliest entry times get the cases first.
4. If there are pending cases, and no lawyers are waiting, they are sent to senior partners and are considered as an opportunity lost.
5. Distribution times, and batch size can change each day, each test case)

Also note, no two lawyers can arrive at the same time.
Overworked Mike started having trouble that he later realized were the timestamps of next day case distribution events and entry timestamps of other lawyers in random order.
Mike decided to take the help of his programmer friend and shared those numbers with you.
You have to help Mike, where you are given the entry times of lawyers and the timestamps of next day case distribution events and entry timestamps of other lawyers in random order.

Assembly 32 Bits K2

```
1  #include <iostream>
2  using namespace std;
3
4  #define N 100000
5  #define M 100000
6
7  #define getint() scanf("%d", &int)
8  int getint()
9  {
10     int x;
11     while (getchar() != '\n')
12         continue;
13     return x;
14 }
15
16 #define getlong() scanf("%ld", &long)
17 long getlong()
18 {
19     long x;
20     while (getchar() != '\n')
21         continue;
22     return x;
23 }
```

ASUS



1h 28m
left



test case)

Also note, no two lawyers can arrive at the same time.
Overworked Mike started having visions that he later realised were the timestamps of next day case distribution events and entry timestamps of other lawyers in random order.

Mike decided to take the help of his programmer friend and shared those numbers with you.

Help

You have to help lazy Mike with the latest time he could reach office without missing out on his case streak for that particular day.

All

Time of case distribution and lawyers office entry time is denoted with integers, starting the day at time 0.

1

Do not expect the distribution schedule times array and lawyer arrival times array to be sorted.

2

Input Format

3

4

- First line consists of number of distribution events for that day
- Next line has space separated timestamps for all events for that day
- Next line has number of lawyers who Mike is competing with
- Next line has space separated timestamps for entry of lawyers
- At last the no. of cases per distribution event is given

Constraints

- n = no. of case distribution events in a day
- m = lawyers competing for the zone
- $1 \leq n, m$ cases per distribution event $\leq 10^4$
- $2 \leq$ distribution event, lawyer entry $\leq 10^4$
- Each element in the distribution timestamps array is unique
- Each element in the lawyer entry timestamps array is unique

Assembly 22 (Mar 4 2)

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     // getting user's input
7     int n, m;
8     cin >> n >> m;
9     // for each distribution event
10    int t[n];
11    int a[m];
12
13    // getting user's input
14    for (int i = 0; i < n; i++)
15        cin >> t[i];
16    for (int i = 0; i < m; i++)
17        cin >> a[i];
18
19    // code for program
20    // for each distribution event
21    for (int i = 0; i < n; i++)
22        // for each lawyer
23        for (int j = 0; j < m; j++)
```

ASUS



1h 28m
left



Output Format

- Print a single timestamp which is the latest Mike could reach the office and get a case assigned successfully.

Sample Input 1

```
3
20 30 10
7
19 13 26 4 25 11 21
2
```

Sample Output 1

```
20
```

Explanation 1

Suppose Mike arrives at timestamp 20,

At timestamp 10 distribution event:
1 case is assigned to the lawyer who arrived at timestamp 4, the
second case is sent to a senior partner/ignored.

At timestamp 20 distribution event:
cases are assigned to lawyers who arrived at timestamp 13, 19

At timestamp 30 distribution event:
Lawyer who arrived at 19, and Mike if arrived at timestamp 20, will
get the cases.

Help

All

1

2

3

4

Assembler 12 (Mar 4/20)

```
1  data
2  userinput
3
4  .text
5  .main
6
7  # getting user's input
8  li $a0, 0
9  la $a1, userinput
10 li $a2, 0
11 syscall
12
13 # display user's input on screen
14 li $a0, 0
15 la $a1, userinput
16 syscall
17
18 # code for program end
19 li $a0, 0
20 syscall
21
```

ASUS



1h 28m
left



Help

All

1

2

3

4

At timestamp 30 distribution event:
Lawyer who arrived at 19, and Mike if arrives at timestamp 20; will
get the cases.

Notice if Mike had entered any later, then the lawyer who entered at
timestamp 21 would have received the case, and Mike would have
lost his streak and bonus.

Sample Input 2

```
2
10 20
4
2 17 18 19
2
```

Sample Output 2

```
16
```

Explanation 2

First distribution event at timestamp 10:
Lawyer entered at 2 timestamp will get a case, and 1 case will be
ignored, as no other lawyer is present.

Second distribution event at timestamp 20:
One case will be given to a lawyer who entered at 17, now since no
two lawyers can have same entry times, Mike could not enter at 19,
19, which means he has to enter office at timestamp 20 to be able to
get a case that day.

Assembler 12 (Mar 4)

```
1  [data]
2  userInput: space 20 & reserved space 20
3
4  test
5  main:
6
7  # getting user's input
8  [[ $0, 0
9  [[ $0, userInput
10 [[ $0, 20
11 syscall
12
13 # display user's input on console
14 [[ $0, 4
15 [[ $0, userInput
16 syscall
17
18 # code for program end
19 [[ $0, 0
20 syscall
21
```

ASUS



1h 27m
left



Help

All

1

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3. Dora The Explorer

Coding

UNSOLVED

Problem Statement

Dora is vacationing on an archipelago of islands, the archipelago has N islands. Dora has decided to start her vacation on Island S and end it on Island E . The archipelago offers travel between islands through a Boat and a Sea Plane, Being quite an explorer Dora wants to travel using only Boat for one part of the vacation and using only Plane for the other part of the vacation. Dora must switch her mode of transportation exactly once.

For example, Dora can travel from Island S to some island I (either directly or through other islands) using the Boat only and then from island I to E (either directly or through other islands) using the Plane only. Alternatively, Dora can travel from Island S to island I using the Plane only and then from island I to E using the Boat only.

Dora has a chart showing the cost of travel from each island to other islands by both Boat and Plane. Help Dora minimize the cost of the vacation. Return the minimum cost, return -1 if it's not possible to complete the journey.

Input Format

- First line contains N , a single integer indicating the number of islands in the archipelago.
- The next N lines contain N space separated numbers. This is a matrix B , where B_{ij} is the cost of travel from island i to island j by Boat.
- The next N lines contain N space separated numbers. This is a matrix P , where P_{ij} is the cost of travel from island i to island j by Plane.

Assembler 32 (Mars 4.5)

```
1  .data
2  userinput: .space 20
3
4  .text
5  main:
6
7  # getting user's input
8  li $v0, 8
9  la $a0, userinput
10 li $a1, 20
11 syscall
12
13 # display user's input
14 li $v0, 4
15 la $a0, userinput
16 syscall
17
18 # code for program
19 li $v0, 10
20 syscall
```




1h 27m
left



Help

All

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4

First line contains N , a single integer indicating the number of islands in the archipelago.

- The next N lines contain N space separated numbers. This is a matrix B , where B_{ij} is the cost of travel from island i to island j by Boat.
- The next N lines contain N space separated numbers. This is a matrix P , where P_{ij} is the cost of travel from island i to island j by Plane.
- The last line contains 2 space separated integers, S and E which are Dora's Start and End islands respectively.

Constraints

- $3 \leq N \leq 1250$
- $1 \leq S, E \leq N$ and $S \neq E$
- $1 \leq B_{ij}, P_{ij} \leq 100$ and $B_{ij}, P_{ij} = -1$ if Dora cannot travel from i to j

Output Format

- A single number indicating the minimum cost of the vacation.

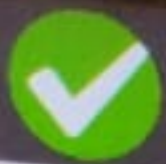
Sample Input 1

```
3
-1 1 2
3 -1 4
5 6 -1
-1 6 5
1 -1 4
3 2 -1
1 2
```

Sample Output 1

Assembler 32 (Mars)

```
1 .data
2 userInput:
3
4 .text
5 main:
6
7 # getting u
8 li $v0, 4
9 la $a0, use
10 li $a1, 20
11 syscall
12
13 # display u
14 li $v0, 4
15 la $a0, use
16 syscall
17
18 # code for j
19 li $v0, 10
20 syscall
```

1h 27m
left



Help

All

1

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4

to j

Output Format

- A single number indicating the minimum cost of the vacation.

Sample Input 1

```
3
-1 1 2
3 -1 4
5 6 -1
-1 6 5
1 -1 4
3 2 -1
1 2
```

Sample Output 1

```
4
```

Explanation

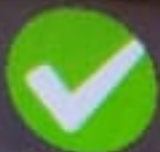
Dora can go from Island 1 to Island 3 by Boat at the cost of 2.
And from Island 3 to Island 2 by Plane at the cost of 2.
Total Cost = 4

Sample Input 2

```
5
-1 3 38 96 1
40 -1 16 24 73
5 52 -1 93 34
78 10 73 -1 84
50 15 51 53 -1
-1 67 30 50 45
96 -1 5 2 12
```

Assembler 12 (M)

```
1 .data
2 userInput:
3
4 .text
5 main:
6
7 # getting
8 li $a0, 8
9 la $a0, userInput
10 li $a1, 20
11 syscall
12
13 # display
14 li $a0, 4
15 la $a0, userInput
16 syscall
17
18 # code for
19 li $a0, 20
20 syscall
```

1h 27m
left



Help

All

1

2

3

4

Sample Output 1

4

Explanation

Dora can go from Island 1 to Island 3 by Boat at the cost of 2.
And from Island 3 to Island 2 by Plane at the cost of 2.
Total Cost = 4

Sample Input 2

5

-1 3 38 96 1

40 -1 16 24 73

5 52 -1 93 34

78 10 73 -1 84

50 15 51 53 -1

-1 67 30 50 45

96 -1 5 2 12

60 19 -1 51 46

-1 56 79 -1 2

83 2 30 5 -1

1 4

Sample Output 2

5

Explanation

Dora can go from Island 1 to Island 5 by Boat at the cost of 1.
And from Island 5 to Island 2 at the cost of 2 and from Island 2 to
Island 4 at the cost of 2 by Plane.

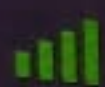
Execution time limit

Assemblies 12 (Mar 4, 2020)

```
1 .data
2 userInput: .space
3
4 .text
5 main:
6
7 # getting user's
8 li $w0, 0
9 la $a0, userInput
10 li $a1, 20
11 syscall
12
13 # display user's
14 li $w0, 4
15 la $a0, userInput
16 syscall
17
18 # code for program
19 li $w0, 10
20 syscall
```

ASUS

1h 27m
left



Help

All

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4. Fun Friday

Coding

UNSOLVED

Problem Statement

As a fun Friday activity, Harsh and Pranay decided to play a new game. The game begins with Harsh giving a list of n numbers to Pranay.

In each iteration,

Pranay adds up all the numbers in the list and adds that sum to his points. Then, he returns the list of numbers to Harsh.

In each iteration, Harsh performs the following actions:

1. If Harsh gets a list with only one number, he throws it out of the game.
2. If Harsh gets a list consisting of more than one number, he divides the list into two non-empty subsequences and gives both subsequences to Pranay one by one.

After Harsh and Pranay complete all iterations, Pranay checks his total points,

determine the maximum possible points Pranay can get.

Input Format

- The first line contains a single integer n ($1 \leq n \leq 10^5$).
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^4$) — the initial numbers that are given to Pranay.

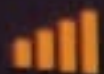
Constraints

- $1 \leq n \leq 10^5$

Assembler 32 (B)

```
1  .data
2  user:in
3
4  .text
5  main:
6
7  # getting
8  li $s0, 0
9  la $a0,
10 li $a1,
11 syscall
12
13 # displa
14 li $s0, 0
15 la $a0,
16 syscall
17
18 # code f
19 li $s0, 0
20 syscall
21
```


1h 26m
left



Help

All

1

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3

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Harsh's Turn:

- Harsh throws the first list (2) out of the game. Then, he divided the second list into two lists, (4) and (6), and returned each list one by one to Pranay.

Pranay's Turn:

- Pranay added (4) to his game points, which are updated to 28.
- Then Pranay added (6) to his game points, updated to 34.
- Finally, he returned both lists to Harsh.

Harsh's Turn:

- Harsh throws both (4) and (6) lists out of the game.

Game Ended.

Sample Input 1

1

11

Sample Output 1

11

Explanation 1

Pranay turn : Add 11 to points updated points = 11;

Harsh turn: Get list (11) throws it out.

Game ended.

Execution time limit

10 seconds

Having an issue with this question? [Report](#)

Assembler 32 (Mars 4.5)

```
1 .data
2 userinput: .space 100
3
4 .text
5 main:
6
7 # getting user's input
8 li $v0, 8
9 la $a0, userinput
10 li $a1, 20
11 syscall
12
13 # display user's input
14 li $v0, 4
15 la $a0, userinput
16 syscall
17
18 # code for program end
19 li $v0, 10
20 syscall
21
```

ASUS

the second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^6$) — the initial numbers that is given to Pranay.

Constraints

- $1 \leq n \leq 3 \cdot 10^5$
- $1 \leq a_i \leq 10^6$

Output Format

- Print a single integer — the maximum possible point.

Sample Input 0

```
3
4 2 6
```

Sample Output 0

```
34
```

Explanation: 0

Pranay's Turn:

- Pranay added the sum of initial numbers ($4+2+6$) to his game points, which is equal to 12. Then, he returned the list to Harsh.

Harsh's Turn:

- Harsh divided the second list into two lists, (2) and (4,6). Then, he returned each list one by one to Pranay.

Pranay's Turn:

- Pranay added 2 to his game points which are updated to 14.
- Then Pranay added (4,6) to his game points, which are updated to 24.

Assembler 32 (Mars 4.5)

```
1 .data
2 userInput: .space 20
3
4 .text
5 main:
6
7 # getting user's input
8 li $v0, 8
9 la $a0, userInput
10 li $a1, 20
11 syscall
12
13 # display user's input
14 li $v0, 4
15 la $a0, userInput
16 syscall
17
18 # code for program end
19 li $v0, 10
20 syscall
21
```

Custom input

ASUS